Building a Python Adventure Game

Course-End Project Report

Project	Text-Based Adventure Game
Language	Python
Development Tool	Visual Studio Code
AI Assistance	GitHub Copilot
Main Script	adventure_game.py

This report documents the development, GitHub Copilot assistance, testing, challenges, and enhancements made to the Python adventure game.

1. Project Objective

The objective was to create an interactive command-line adventure game in which the player explores an ancient land, makes decisions, encounters different outcomes, and attempts to find a legendary treasure. The project reinforces fundamental Python concepts such as variables, input/output, functions, conditionals, and loops.

2. Tools and Technologies

Visual Studio Code: Used to create, edit, save, and execute the Python program.
Python: Used to implement the interactive CLI and game logic.
GitHub Copilot: Used to assist with generating the main game functions and replay loop.

3. Game Structure and Features

Component	Implementation
start_game()	Welcomes the player, asks for their name, and presents forest/cave choices.
forest_path()	Provides river or tree choices with winning/losing outcomes.
cave_path()	Provides torch or darkness choices with winning/losing outcomes.
main()	Runs the game in a while loop and provides replay functionality.
Game outcomes	The player can find the treasure, lose the adventure, or restart.

4. How GitHub Copilot Assisted

GitHub Copilot was used to generate and refine individual parts of the game, including **start_game()**, **forest_path()**, **cave_path()**, and the main replay loop. Early suggestions did not fully match the project requirements, so more specific prompts were provided. The suggestions were reviewed and adapted before being accepted. This showed how AI assistance can speed up coding while the programmer remains responsible for checking correctness and requirements.

5. Challenges Faced

The main challenge was ensuring that AI-generated code followed the exact problem statement. An early suggestion introduced unrelated directional choices, so the prompt was refined to require the dark forest and mysterious cave structure. The completed program was then tested across multiple decision paths.

6. Enhancements and Modifications

The game was kept beginner-friendly while adding a personalized player name, two routes toward the treasure, clear winning and game-over messages, and a replay option. The successful forest river route was also extended so that it leads to a hidden cave containing the legendary treasure.

7. Testing and Sample Outputs

The completed program was tested using the main decision paths. The following screenshots are actual terminal outputs from the completed game.

Test 1 — Forest → Follow the river: successful treasure discovery.

```
PS C:\Users\yarra\OneDrive\Desktop\Python_Adventure_Game> & "C:\Program Files\Python311\python.exe" c:/Users/yarra/OneDri
ve/Desktop/Python_Adventure_Game/adventure_game.py
Adventure Game Setup Succesfull
Welcome, adventurer!
You are on a quest to find the legendary treasure.
What is your name, brave explorer? Raghav
Good luck, Raghav!
You have two paths ahead:
1. Explore the dark forest
2. Enter the mysterious cave
Enter 1 or 2 to choose your path: 1
You step into the dark forest, searching for the legendary treasure.
You have two choices:
1. Follow the river
2. Climb a tree
Enter 1 or 2 to choose your path: 1
You follow the river, and after a long journey, you discover a hidden cave.
Inside the cave, you find the legendary treasure shining brightly!
Congratulations, brave explorer! You have won the game!
```

Test 2 — Forest → Climb a tree: game-over outcome.

```
Welcome, adventurer!
You are on a quest to find the legendary treasure.
What is your name, brave explorer? Raghav
Good luck, Raghav!
You have two paths ahead:
1. Explore the dark forest
2. Enter the mysterious cave
Enter 1 or 2 to choose your path: 1
You step into the dark forest, searching for the legendary treasure.
You have two choices:
1. Follow the river
2. Climb a tree
Enter 1 or 2 to choose your path: 2
You climb the tree, but slip and fall. Game over!
```

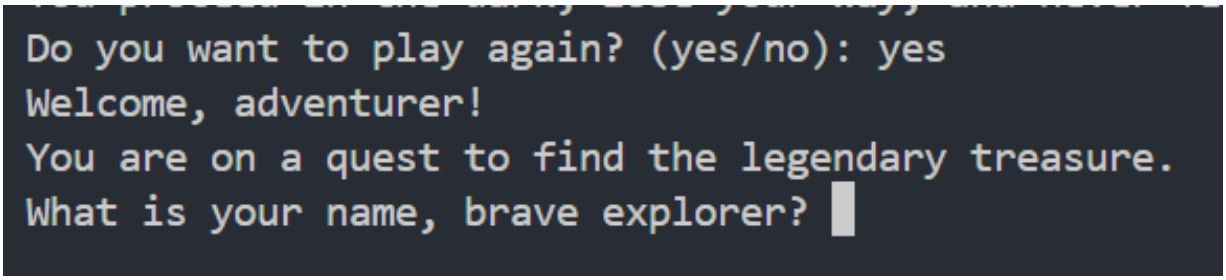
Test 3 — Cave → Light a torch: successful treasure discovery.

```
Do you want to play again? (yes/no): yes
Welcome, adventurer!
You are on a quest to find the legendary treasure.
What is your name, brave explorer? Raghav
Good luck, Raghav!
You have two paths ahead:
1. Explore the dark forest
2. Enter the mysterious cave
Enter 1 or 2 to choose your path: 2
You enter the mysterious cave, searching for the legendary treasure.
You have two choices:
1. Light a torch
2. Proceed in the dark
Enter 1 or 2 to choose your path: 1
You light the torch, and the cave walls glow to reveal the legendary treasure. You win!
```

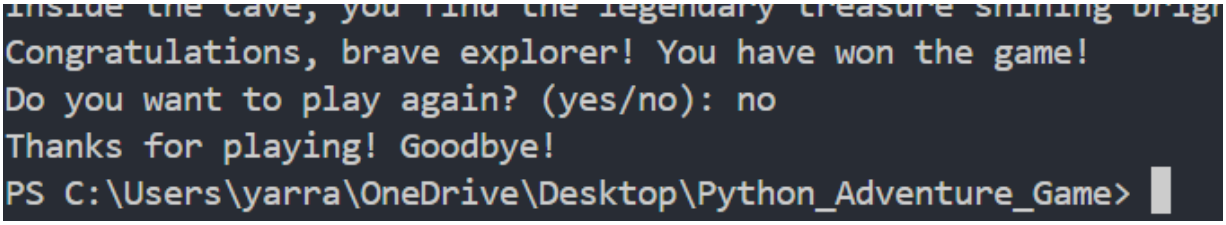
Test 4 — Cave → Proceed in the dark: game-over outcome.

```
Welcome, adventurer!
You are on a quest to find the legendary treasure.
What is your name, brave explorer? Raghav
Good luck, Raghav!
You have two paths ahead:
1. Explore the dark forest
2. Enter the mysterious cave
Enter 1 or 2 to choose your path: 2
You enter the mysterious cave, searching for the legendary treasure.
You have two choices:
1. Light a torch
2. Proceed in the dark
Enter 1 or 2 to choose your path: 2
You proceed in the dark, lose your way, and never find the treasure. Game over!
```

Test 5 — Replay option: entering yes starts a new adventure.



Test 6 — Replay option: entering no exits with a goodbye message.



8. Conclusion

The Python Adventure Game was successfully implemented as an interactive command-line application. The final program demonstrates functions, conditionals, input/output, and a while loop, while providing multiple decision-based outcomes and a restart facility. GitHub Copilot assisted with code generation, and the generated code was reviewed and modified to match the assignment requirements. The game was tested across winning, losing, and replay scenarios.